

BIOL 4559 Homework 03

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Question 1.

The configuration model is a useful tool for matching degree heterogeneity while randomizing all other structure. I've attached code snippets in a Jupyter notebook that will be helpful in completing some of the assignment.

- (a) Read up on the network configuration model in Coscia Chapter 18.1.

Solution Attempt Done!

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- (b) Code up an implementation of the network configuration model by doing the following

- Write a function called `configuration_model` which accepts a list (or array) of degrees as input and returns an edge list.

Solution Attempt In notebook!

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- Check that the sum of the degrees is even and expand the list of degrees into a list of nodes, where the number of times that a node appears in this list is its degree.

Solution Attempt Implemented!

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- Form an edge list by randomly permuting the list of nodes with (`numpy.random.shuffle`) and then pairing subsequent nodes (index 0 with index 1, index 2 with index 3, etc.)

Solution Attempt Implemented!

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- Add an additional argument called `return_graph` to `configuration_model` and use this argument to allow the function to return either an edge list or a network as a NetworkX MultiGraph object.

Solution Attempt Implemented!

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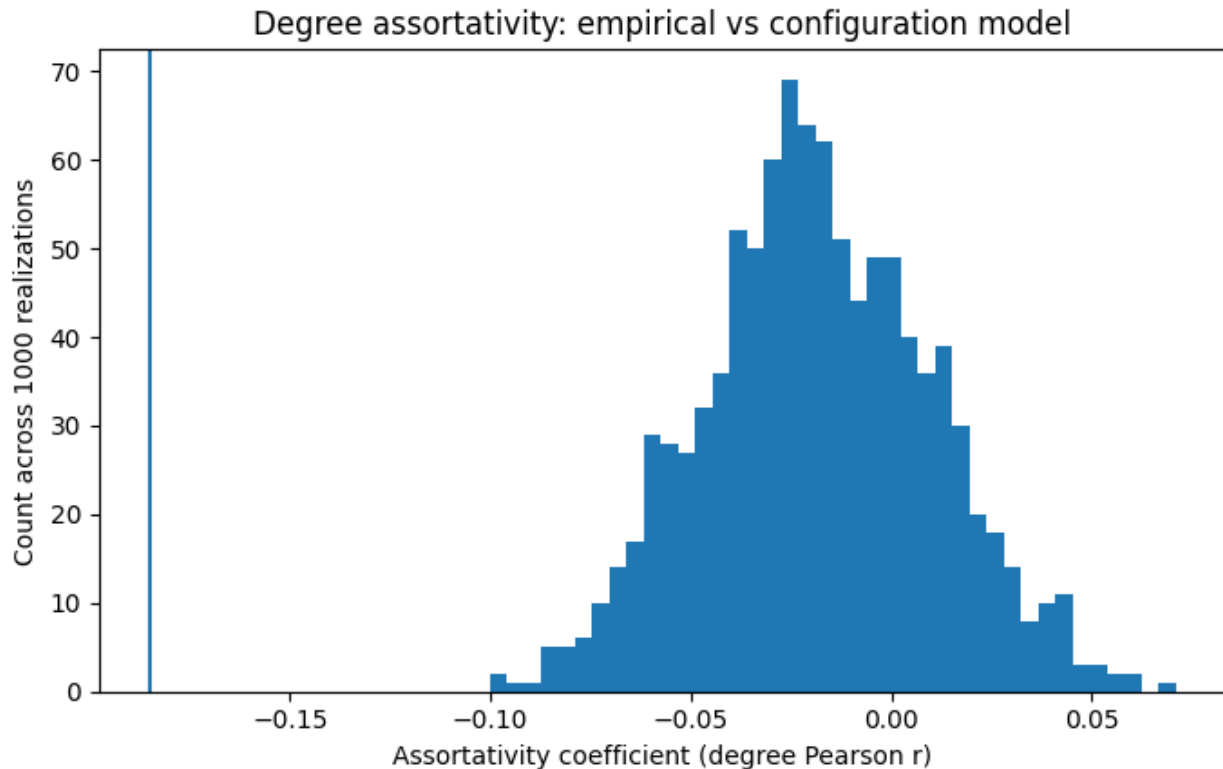
- (c) Generate a degree distribution from the truncated power law distribution (see attached code snippet) with `N=1000`, `minval=1`, `maxval=N-1`, `alpha=-2.5`. If the sum of degrees is odd, add one to a random node. Input this degree sequence into your function and output an edge list. Are there any self-loops or multi-edges in the resulting edge list? If so, roughly what fraction of the total number of edges are multi-edges and self-loops, respectively? Average over 1,000 realizations of the configuration model. How does this compare to the `budapest_connectome` dataset (on Canvas)?

Solution Attempt In notebook!

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- (d) Import the edge list of the budapest connectome dataset (on Canvas). (i) Create a NetworkX MultiGraph object from this edge list, (ii) generate 1,000 realizations of the configuration model fitted to this edgelist, and (iii) plot a vertical line indicating the degree assortativity (this is implemented in NetworkX) of the empirical dataset and a histogram of degree assortativity values computed from the ensemble of configuration model networks. Make sure that the axes are labeled! Is the degree assortativity more or less than you'd expect at random?

Solution Attempt Empirical assortativity: -0.185, Ensemble mean: -0.018, Ensemble std: 0.028.



The empirical assortativity of -0.185 is much lower than the ensemble mean of -0.018, which suggests it is more disassortative than expected at random. This means high-degree nodes are less likely to connect with each other than chance would predict.

□

Question 2.

The Medici family was a powerful political dynasty and banking family in 15th century Florence. The classic network-based explanation of their power, offered by Padgett and Ansell in 1993, claims that they established themselves as the most central players within the network of prominent Florentine families (shown in Fig. 1).

- (a) Construct a table with four columns, each containing a list of (family, score) pairs for the following centrality score functions: degree centrality, harmonic centrality, eigenvector centrality, betweenness centrality. All of these are implemented in NetworkX. Within each centrality, sort the pairs in decreasing order of importance; three decimal places is sufficient detail. Discuss (i) the degree to which these scores agree with Padgett and Ansell's claim that the Medicis occupied a structurally important position in this network, and (ii) what the scores say about the second most important family

Solution Attempt The centrality results back Padgett and Ansell: Medici sits first on degree, eigenvector, betweenness, and harmonic, which means they had the most ties, were linked to other well connected families, controlled many shortest paths, and could reach others quickly. For the runner-up, it depends on what we value. Strozzi looks second by degree and eigenvector, signaling a strong hub tied to other prominent houses. Guadagni ranks second by betweenness and harmonic, pointing to a family with good brokerage power and wide reach across the network.

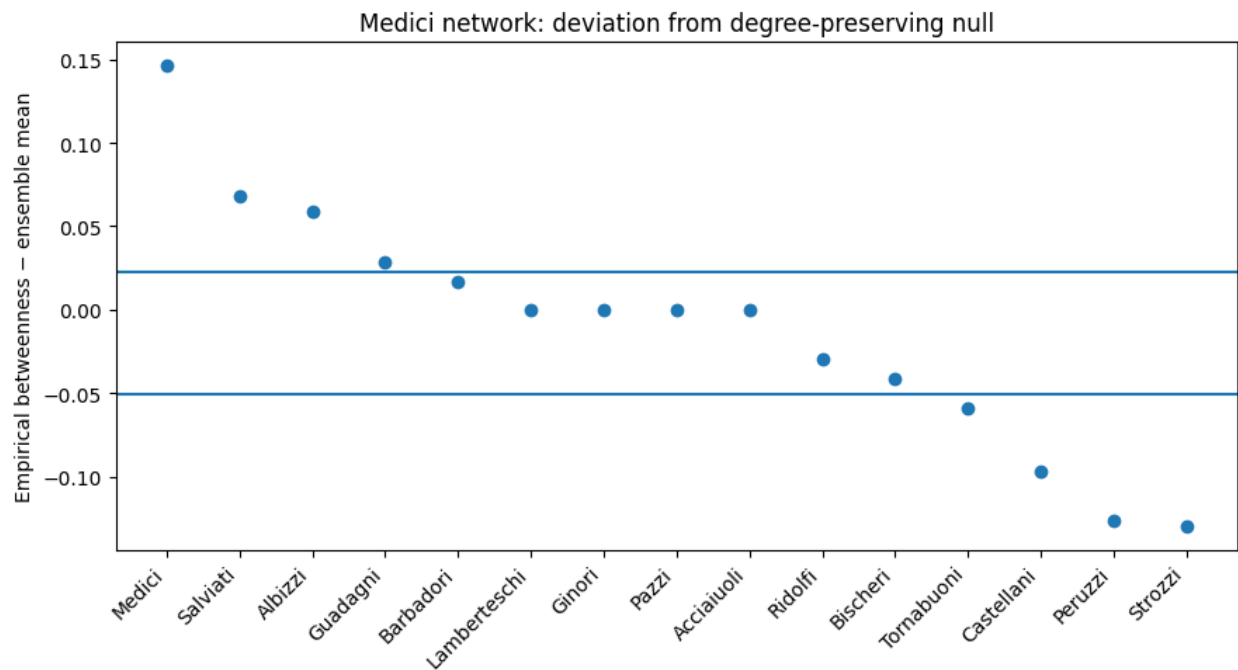
Family	Degree	Harmonic	Eigenvector	Betweenness
Medici	0.429	9.500	0.430	0.522
Strozzi	0.286	8.083	0.356	0.255
Guadagni	0.286	8.000	0.342	0.212
Castellani	0.214	7.833	0.326	0.143
Peruzzi	0.214	7.833	0.289	0.114
...	...	...	...	...
Salviati	0.143	6.583	0.146	0.022
Acciaiuoli	0.071	5.917	0.132	0.000
Pazzi	0.071	5.367	0.089	0.000
Ginori	0.071	5.333	0.075	0.000
Lamberteschi	0.071	4.767	0.045	0.000

- (b) Determine whether the relative structural importance of the Medici can be explained solely by the degree sequence of the network G . To do this, use the configuration model you implemented prior to produce an ensemble of 1,000 random graphs with the same degree sequence d as G , and extract a distribution of betweenness centrality scores for each node.

Make a figure showing the difference between a node's betweenness centrality on G and its average betweenness centrality in the ensemble. On this figure, include lines showing the 25% and 75% quantiles of the distribution around the mean for all vertices (as in the figure below, for the Karate club).

Finally, (i) discuss your results here in terms of Padgett and Ansell's story of the Medici family (see article on Canvas), and (ii) comment on whether your results from part a) have changed.

Solution Attempt Medici delta (empirical mean): 0.147; Top 5 by positive delta: ['Medici', 'Salviati', 'Albizzi', 'Guadagni', 'Barbadori']; 25% and 75% quantile lines: -0.051 0.023. *Clearer results in the notebook.*



If Medici's delta is well above the 75% line and near the top of the list, their high betweenness is not just a by-product of degree; it reflects brokerage across factions, consistent with Padgett Ansell's story. This doesn't change the conclusion from part (a): Medici remain structurally dominant. If another house (often Guadagni) also shows a large positive delta, note their brokerage role as a strong runner-up.